Hide menu

Week 4: Appearance Matching

- **Ungraded Plugin:** 4.1 Overview of Appearance Matching — 4 min
- **Ungraded Plugin:** 4.2 Shape vs. Appearance — 6 min
- **Practice Assignment:** 4.2 Shape vs. Appearance Self-Check Quiz — Submitted
- **Ungraded Plugin:** 4.3 Learning Appearance — 6 min
- **Practice Assignment:** 4.3 Learning Appearance Self-Check Quiz — Submitted
- **Ungraded Plugin:** 4.4 Principal Component Analysis — 17 min
- **Practice Assignment:** 4.4 Principal Component Analysis Self-Check Quiz — Submitted
- **Ungraded Plugin:** 4.5 Finding Principal Components — 8 min
- **Practice Assignment:** 4.5 Finding Principal Components Self-Check Quiz — 3h
- **Ungraded Plugin:** 4.6 PCA and SVD — 7 min
- **Practice Assignment:** 4.6 PCA and SVD Self-Check Quiz — 3h
- **Ungraded Plugin:** 4.7 Parametric Appearance Representation — 12 min
- **Practice Assignment:** 4.7 Parametric Appearance Representation Self-Check Quiz — 3h
- **Ungraded Plugin:** 4.8 Appearance Matching — 14 min
- **Practice Assignment:** 4.8 Appearance Matching Self-Check Quiz — 3h
- **Discussion Prompt:** Week 4 Smartphones using 3D Sensors — 10 min

Week 4 Quiz

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

4.4 Principal Component Analysis Self-Check Quiz

1. Consider a set of points in a 2D space. In which of the following cases does dimensionality reduction with PCA not provide any additional benefit? **1 / 1 point**

- The set of points lie on a straight line
- The set of points lie on a circle
- The set of points are uncorrelated
- None of the above

Correct
In the case of a straight line and a circle, utilizing PCA certainly helps to reduce them to 1D and 2D, respectively. However, if the set of points is already uncorrelated, then the principal components will not capture any variance or information that is not already present in the original features. In short, dimensionality reduction with PCA does not provide any additional benefit in such a situation.

2. Which of following conditions, if satisfied, ensures that the backward projection $\sum_K p_k e_k = f$? **1 / 1 point**

- K < N
- K = N
- Both A and B
- None of the above

Correct
K = N simply means that we are using all the dimensions of the original subspace, so they capture all the variance in the data, resulting in a reconstruction/backward projection error of 0.

Help me practice
Let's chat
Assignment details
Submitted
Attempt 1
Try again
Your grade
To pass you need at least 80%. We keep your highest score.
100%
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